

Computational Financial Modelling Lab: 1

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Course: Computational Financial Modeling Lab

Index Focus: IBEX-35 (Spain)

Time Horizon: 2010-2025

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1. Introduction

This report concludes my assignment-1 of the course `Computational Financial Modeling`. I have chosen to work on `IBEX-35`, Spain and this report have all the work and analysis related to IBEX-35 index only else stated otherwise.

2. Data Sources

To keep things reproducible and simple, I extracted data from simple places:

- **Yahoo Finance:** It's free, accessible, and has historical data going back to 2010.
- **IBEX-35 Constituent Lists:** I gathered biannual snapshots (June and December) of the index's companies from 2010 to 2025. These came from official sources and were compiled into CSVs for analysis.
- **Wikipedia:** All data has been verified manually from the wikipedia's history tab.
- **Custom Mapping Files:** I created these myself to handle messy data—things like standardizing company names (e.g., fixing variations like "Banco Santander" vs. "Santander") and assigning sectors (e.g., utilities, finance).

Assumptions: Prices are adjusted for dividends and splits where possible, and I focused on tradable stocks only.

3. Folder Structure & Data Organization

I tried to keep things simple and inplace and here is the breakdown of the files which you will be encountering during your work:

Root Directory Files

- **1-Q1.txt**: Answers and explanations for Question 1—intro concepts, background on indices, and why they matter.
- **2-Q2.txt**: Written analysis for Question 2, comparing indices and reasoning through sector influences.
- **3-Q3.txt**: Observations for Question 3, including how indices handled the 2022 global slowdown.
- **4-IBEX35Constituents-10-25.xlsx**: An Excel file with the full history of IBEX-35 companies from 2010–2025. Great for manual checks.
- **5-Q5.py**: Code for the Question 5 of assignment to get data based on user inputs and create the final master lasting stocks.
- **6-Q6.py**: Code for the Question 6 of assignment to pull the dividends and splits of the main lasting stocks.
- **Buildmaster.py**: Code to combine individual stock CSVs into one master price database (Helper Code).
- **SplitsDivCombo.csv**: Summarizes corporate events (splits and dividends).
- **SplitsDividend.xlsx**: The Excel version of the above for easy viewing.
- **Report.pdf**: This Report.
- **Struct.pdf**: File explaining whole folder struct.

Note: Read the file `Struct.pdf` to get full idea of the File Structure.

4. Index Selection and Major Sector

4.1 Index Rebalancing Cycles: A Comparative Overview

Index	Rebalancing Frequency	Effective Dates	Announcement Lead Time
NSE-50 (India)	Semi-Annually	Last trading day of March and September	Approximately 1 month in advance

Index	Rebalancing Frequency	Effective Dates	Announcement Lead Time
Sensex (India)	Semi-Annually	Monday after the 3rd Friday of June & December	Approximately 1 month in advance
IBEX-35 (Spain)	Semi-Annually	First trading day after the 3rd Friday of June & December	Approximately 1 month in advance
OMXS-30 (Sweden)	Semi-Annually	First trading day of January & July	At least 5 trading days in advance
ATX-20 (Austria)	Semi-Annually	March & September	Not explicitly stated; up to 3 constituent changes per review

Selected Index: IBEX-35 (Spain)

4.2 Why IBEX-35?

- **Reorganizing frequency:** Twice a year (June & December)
- **Transparency:** Changes announced ~1 month in advance
- **Consistency:** Regular, predictable review cycle aligned with European market practices
- **Sector diversity:** Well-balanced across traditional and modern industries

4.3 Sectoral Composition of IBEX-35 (Approximate Weights)

Sector Category	Approximate Weight	Key Characteristics
Financials (Banks & Insurance)	30–35%	Dominant sector; includes major banks like Santander, BBVA
Utilities & Energy	20–25%	Stable, regulated cash flows; includes Iberdrola, Endesa
Industries & Infrastructure	15–20%	Cyclical exposure; includes Ferrovial, ACS
Consumer & Retail	10–15%	Includes global brands like Inditex (Zara)
Telecom & Media	5–8%	Moderately weighted; includes Telefónica
Pharma & Health	3–4%	Smaller but growing presence; includes Grifols

4.4 Key Observations on IBEX-35 Structure

1. **Financials are the backbone** – Constituting around one-third of the index, reflecting Spain’s banking-heavy economy.

2. **Utilities & Energy provide stability** – High weight in regulated sectors adds defensive characteristics.
3. **Cyclical sectors are present but moderate** – Industries and consumer sectors balance growth and risk.
4. **Technology underweight** – Unlike US indices, tech is not a dominant sector in IBEX-35.

4.5 Conclusion & Relevance to Assignment

The **IBEX-35** was chosen for this analysis due to its:

- **Predictable rebalancing schedule** (June & December)
- **Transparent and consistent review process**
- **Sectoral diversity** – allowing for meaningful survival and sectoral analysis
- **Significant weighting in stable, regulated industries** – ideal for studying long-term survivorship

This structural understanding sets the foundation for the **survivor analysis, sector mapping, and corporate action tracking** performed in the later stages of the assignment.

5. Comparative Index Selection: Why IBEX-35?

5.1 Sector-Based Comparison of Indices

When selecting a European index to compare with Indian benchmarks, I evaluated several based on **sectoral composition, market depth, and economic exposure**. Here's how they stack up:

IBEX-35 (Spain)

- **Major Sectors:** Financials (Banks), Utilities & Energy
- **Economic Profile:** Infrastructure-heavy, domestic + regional (EU/LATAM) exposure
- **Market Size:** Large, liquid, and institutionally tracked
- **Why it resonates with Indian indices:**

Like India's Nifty 50 and Sensex, the IBEX-35 is **dominated by Financials and Energy/Utilities**—sectors that are highly regulated, domestically anchored, and resilient to global trade volatility.

Indian Indices (Nifty 50 & Sensex)

- **Major Sectors:** Financials (Banks & NBFCs), Energy & Utilities, IT Services
- **Economic Profile:** Large domestic consumption, moderate export exposure (IT, pharma)
- **Market Size:** Large, diverse, and globally significant
- **Key Similarity with IBEX-35:**

Both are **domestically-driven economies** with strong banking sectors and significant

utility/energy representation, making them structurally comparable.

OMXS-30 (Sweden)

- **Major Sectors:** Industrials & Manufacturing, Export-oriented global firms
- **Economic Profile:** Highly export-dependent, sensitive to global trade cycles and forex fluctuations
- **Why it was not chosen:**
The OMXS-30 is **structurally different** from Indian indices. Its heavy reliance on exports makes it more volatile during global slowdowns, unlike the domestically-anchored Indian and Spanish markets.

ATX (Austria)

- **Major Sectors:** Financials, Energy & Utilities
- **Economic Profile:** Highly concentrated, small market, regional exposure to Central & Eastern Europe
- **Why it was not chosen:**
While the ATX's sector mix somewhat mirrors India's, it is **too small and illiquid** for a fair comparison. The IBEX-35 offers better volume, diversity, and institutional relevance.

5.2 Final Selection: IBEX-35

Criterion	IBEX-35	ATX	OMXS-30
Sector Match with India	✓ High	✓ Moderate	✗ Low
Market Liquidity	✓ High	✗ Low	✓ High
Domestic/Export Balance	✓ Domestic-heavy	✓ Mixed	✗ Export-heavy
Index Size & Relevance	✓ Large, tracked	✗ Small, niche	✓ Large, but misaligned

Decision:

IBEX-35 was chosen because it:

- Shares a **similar sectoral footprint** with Indian indices (Financials + Energy/Utilities)
- Has **comparable market depth and liquidity**
- Is **domestically-driven**, making it a better parallel for studying resilience during global slowdowns

6. Global Market Slowdown 2022 – Index Comparison

6.1 Objective

Compare how five major stock indices performed during the 2022 global slowdown.

We looked at:

- **Nifty 50** and **Sensex** (India)
- **IBEX 35** (Spain)
- **OMXS 30** (Sweden)
- **ATX** (Austria)

All indices were normalized to 100 as of January 2022 for fair comparison.

6.2 Overall Performance

- **Indian indices (Nifty, Sensex)** held up best, ending near or above starting levels.
- **European indices** fell more, especially **ATX** (Austria) and **OMXS 30** (Sweden).
- **IBEX 35** was in the middle—down but not as sharply.

6.3 Drawdowns (Peak to Trough)

- **ATX** dropped almost **-30%** at its worst.
- **OMXS 30** fell **-20 to -25%**.
- **IBEX 35** declined **-10 to -15%**.
- **Nifty & Sensex** saw only **0 to -10%** drawdowns.

Takeaway: European markets felt the pain much more than India.

6.4 Relative Performance vs. Nifty 50

- **Sensex** moved almost exactly like Nifty.
- **IBEX 35** underperformed by **-7 points** by early 2023.
- **OMXS 30** underperformed by **-18 points** at its worst.
- **ATX** was the weakest, underperforming by **-22 points**.

6.5 Why Did Europe Fall More?

Geopolitical Shock

- **Ukraine War** hit Europe directly—energy crisis, inflation, and uncertainty.
- India was far away and less impacted.

Sector Differences

- **India:** Heavy in IT, pharma, consumer goods—domestic-driven.

- **Europe (ATX, OMXS):** More cyclical, export-focused, energy-sensitive industries.

Monetary Policy

- **Europe:** Aggressive rate hikes to fight inflation.
- **India:** More measured tightening, softer on markets.

Currency & Flows

- European currencies weakened more than the Indian rupee.
- Foreign investors pulled more money from Europe than India.

6.6 Conclusion

- **European indices** fell harder due to **war exposure, energy reliance, and tighter policy**.
 - **Indian indices** were more resilient thanks to **Domestic focus, IT industries and less direct shock**.
 - **IBEX 35** was somewhere in between—hurt but not crushed.
 - For comparative analysis see the graphs in the main directory.
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7. IBEX-35 Master Constituent File

I created the **exact composition of the IBEX-35 index over the last 15 years**, broken down **biannually** (every 6 months).

7.1 File Created:

`4-IBEX35Constituents-10-25.xlsx`

This is the **master Excel file** containing the full 15-year constituent history.

7.2 File Structure:

- **30 sheets** – one for each biannual snapshot
- **Sheet naming convention:** `Jun-2010` , `Dec-2010` , `Jun-2011` , ..., `Dec-2024`
- Each sheet contains the **list of companies** that were part of the IBEX-35 during that 6-month period

7.3 How It Was Built:

1. Used **biannual constituent snapshots** (June & December) from 2010 to 2025
2. Verified each list against official historical data

3. Cleaned company names to ensure consistency across sheets
4. Compiled into a single, structured Excel workbook
5. Scrapped Wikipedia for verification too.

7.4 What's Inside Each Sheet:

- **Company names**
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8. Generic Survivor Analysis

8.1 Overview

A **flexible Python program** that:

1. **Takes user input** for either:
 - Number of years (1–15) **or**
 - A specific date range (2010–2025, biannually)
2. **Identifies survivors** (companies that stayed in the IBEX-35 throughout the period)
3. **Maps each survivor to its sector**
4. **Creates a restricted price database** for those survivors
5. **Analyzes which sector contributed the most survivors**

8.2 How It Works

Step 1: User Chooses Analysis Mode

The user selects:

- **Option 1:** Number of years (e.g., 10 years)
 - Program finds survivors in **first N years** and **last N years**
- **Option 2:** Custom date range (e.g., Jun-2012 to Dec-2019)
 - Finds survivors **only in that specific window**

Step 2: Loading & Mapping Data

The script:

- Loads biannual constituent lists (`ibex_YYYY_Jun.csv` , `ibex_YYYY_Dec.csv`)
- Uses **canonical name mapping** to avoid duplicate company names
- Maps each company to its **sector** using a sector mapping file

Step 3: Survivor Detection

- Finds companies present in **every snapshot** within the chosen period
- Outputs:
 - Total number of survivors
 - List of companies with sectors
 - Sector-wise survivor counts

Step 4: Price Database Creation

- For each survivor, loads its daily price data from `IBEX35/prices/`
- Filters prices to match the chosen date range
- Saves filtered CSV files in `IBEX35/results/[label]/`

Step 5: Sector Analysis

- Counts survivors per sector
- Identifies the **sector with the most survivors**

8.3 15-Years Survivors Company

Company	Sector
ACS	Infrastructure & Construction
Acciona	Infrastructure & Renewable Energy
ArcelorMittal	Metals & Mining
BBVA	Banking
Banco Sabadell	Banking
Banco Santander	Banking
Bankinter	Banking
Enagás	Gas Transmission & Infrastructure
Ferrovial	Infrastructure & Construction
Grifols	Healthcare & Biotechnology
Iberdrola	Utilities - Electricity
Inditex	Retail & Apparel
Indra Sistemas	Technology & Defense Systems
International Airlines Group	Airlines

Company	Sector
Mapfre	Insurance
Naturgy	Utilities-Gas & Electricity
Red Eléctrica de España	Electricity Transmission
Repsol	Energy - Oil & Gas
Telefónica	Telecommunications

8.4 15-Years Sectors Dominance

Sector	Survivor Count
Banking	4
Utilities - Electricity	2
Energy - Oil & Gas	2
Infrastructure & Construction	2
Airlines	1
Electricity Transmission	1
Healthcare & Biotechnology	1
Infrastructure & Renewable Energy	1
Insurance	1
Metals & Mining	1
Retail & Apparel	1
Technology & Defense Systems	1
Telecommunications	1

9. Corporate Action Tracking for 15-Year Legends

9.1 Objective

Track all **stock splits and bonus issues** for companies that survived in the IBEX-35 for the full 15-year period (2010–2025).

9.2 How It Works

Identify the Survivors

Only companies in the **"15-Year Legends" list** are analyzed:

- 19 companies (e.g., Banco Santander, Iberdrola, Inditex, etc.)
- Each mapped to its Yahoo Finance ticker

Data Extraction

Using the `yfinance` library, the script:

- Downloads **corporate action history** for each stock
- Filters data from **2010 onward**
- Identifies:
 - **Stock Splits** (e.g., 2:1, 5:1)
 - **Bonus Issues / Scrip Dividends** (treated as dividends in Yahoo data)

Categorization

- **Event Type:** "Stock Split" or "Dividend/Bonus Issue"
- **Ratio/Value:** The split ratio or dividend amount
- **Ex-Date:** Date the action took effect

Output Files

- `SplitsDividend.xlsx`:
 - Sheet 1: `Splits` and `Bonus` – All split events
 - Sheet 2: `Dividends` – Dividend/bonus events
- `SplitsDivCombo.csv` – Combined data for further analysis

9.3 Split Table

Company	Ex-Date	Value/Ratio
ACS	2024-01-19	1.011627907
ArcelorMittal	2017-05-18	0.3333333333
Banco Sabadell	2010-04-23	1.018867925
Banco Sabadell	2012-06-01	1.035531552

Company	Ex-Date	Value/Ratio
Banco Sabadell	2013-03-27	1.012935774
Banco Sabadell	2014-03-28	1.008984726
Banco Sabadell	2015-06-08	1.004275697
Banco Sabadell	2016-04-07	1.012012012
Banco Sabadell	2019-12-20	1.015104901
Banco Santander	2011-01-17	1.015384615
Banco Santander	2020-11-16	1.043478261
Bankinter	2013-04-04	1.555555556
Grifols	2016-01-04	2
Iberdrola	2026-01-12	1.01369863
Inditex	2014-07-28	5
Red Eléctrica de España	2016-07-11	4

Interpretation Notes:

- **Value/Ratio > 1** = Forward split (e.g., 2:1, 5:1)
- **Value/Ratio < 1** = Reverse split (e.g., ArcelorMittal's ~1:3 consolidation)
- **Values close to 1** (e.g., 1.0116) are often **scrip dividends or accounting adjustments**, not true splits
- **True splits** are rare: Grifols (2:1), Inditex (5:1), Red Eléctrica (4:1)

9.4 Dividend Table

It has much more data so see in the file [SplitsDividend.xlsx](#).
